

Resource Allocation Optimization in a Food Bank

Project Report: Memetic Algorithm and Preference Reasoning

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Abstract

This report illustrates the development of an Artificial Intelligence system designed to solve the *Food Bank Resource Allocation Problem* (FBRAP). The system employs a memetic approach that hybridizes a Genetic Algorithm (for global optimization) with a greedy post-processing heuristic. Advanced *Preference Reasoning* techniques for priority management and stochastic repair mechanisms based on Softmax distributions have been integrated to maximize fairness and distributional effectiveness in resource-scarce scenarios.

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1 Introduction and Problem Definition

This project addresses the logistical and humanitarian challenge of distributing limited food resources to a network of beneficiary entities (hospitals, canteens, family homes). The objective is not merely an equal distribution, but the maximization of global utility, ensuring that facilities with higher priority and critical needs are satisfied first, while strictly adhering to hard inventory constraints.

2 Data Modeling and Heuristic Priorities

The environment is modeled through three main entities retrieved from a Supabase database: *Beneficiary*, *Person*, and *Product*.

Prior to optimization, the system performs a heuristic calculation of priorities. For every single person, a score is calculated based on age, type of hosting facility, caloric requirement, and gender. Subsequently, the global priority of an Entity E is defined as:

$$Priority(E) = \left(\frac{\sum_{p \in E} priority(p)}{|E|} \times w_1 \right) + (|E| \times w_2) \quad (1)$$

This metric guides the entire Artificial Intelligence, allowing it to apply *Preference Reasoning* in allocation decisions.

3 The Genetic Algorithm and Fitness Evaluation

The core of the search engine is a Genetic Algorithm. Each chromosome represents a complete mapping of allocations (Entity $\rightarrow$ Product $\rightarrow$ Quantity).

3.1 The Evaluation Function (Penalty Method and Threshold)

The fitness score starts from an ideal value (1,000,000) from which penalties are subtracted. The architecture includes:

- **Hard Constraints:** Exceeding physical warehouse stocks incurs massive penalties.
- **Soft Constraints (Nutritional):** Limits on sugars ($< 50g$), salts ($< 6g$), and minimum proteins ($> 60g$).

Crucial for solving the FB RAP is the implementation of the **Minimum Threshold (α -threshold)**, set at 20% of the caloric requirement. The algorithm evaluates three scenarios for each entity:

1. **Unserved (0 kcal):** An **unserved** penalty is applied, multiplied by the entity's priority.
2. **Below Threshold ($< 20\%$):** The **unserved** penalty is applied, added to a **wasted_food** penalty. This inhibits the ineffective dispersion of resources (the so-called "sprinkling" effect).
3. **Served ($\geq 20\%$):** The major penalties are zeroed out, and only a *soft constraint* is applied to push the AI towards fulfilling 100% of the requirement.

4 Smart Repair with Softmax Distribution

During mutations or initialization, solutions may generate physically impossible distributions (inventory excess). Instead of discarding these chromosomes, a smart *Repair* operator was designed.

Instead of subtracting excess food in a purely random manner, the algorithm selects the "target" beneficiary following a probabilistic **Softmax** distribution. The probability weight for each entity i is calculated as:

$$W_i = \exp \left(-P_i \times \frac{F_i^{missing}}{F_i^{total}} \right) \quad (2)$$

Where P_i is the priority of the entity, and the ratio represents the percentage of the requirement yet to be fulfilled. Since the exponent is strictly non-positive, $W_i \in (0, 1]$. Entities with **low priority** and **already widely satisfied requirements** generate an exponent close to zero, maximizing their weight (close to 1) and becoming the preferential targets from which the AI will subtract products. This guarantees an adaptive and urgency-driven conflict resolution.

5 Memetic Hybridization (Greedy Clean-up)

Once the genetic evolution is complete, the optimal solution might leave small residual stocks in the warehouse. The algorithm transforms into a **Memetic Algorithm** by applying a purely deterministic *Local Search / Post-processing* phase. The algorithm iterates through the remnants and assigns them in descending order of priority to the entities that still exhibit a caloric gap, introducing a logical *Hard Constraint*: it is strictly forbidden to exceed 100% of an entity's requirement, effectively preventing physiological waste.

6 Results Analysis

The execution of the system generates analytical metrics and reports:

- **Convergence Plot:** Demonstrates the correct optimization of fitness over the generations.
- **Distributional Fairness and Satisfaction:** The charts highlight the *Triage Effect* operated by the AI: in cases of severe scarcity, the algorithm rapidly saturates the highest-priority facilities, partially sacrificing lower-priority entities to avoid the heavy penalties of *Preference Reasoning*.
- **Tabular Reports:** JSON and CSV exports of the exact allocations, mapped at both the Entity and individual Person levels, ready for use in real-world operational scenarios.

7 Conclusions

The developed model demonstrates how combining Genetic Algorithms, local heuristics, and Preference Reasoning (via penalty functions and Softmax) provides a robust, scalable,

and ethically-oriented solution to complex resource allocation problems in humanitarian logistics.